Reinforcement Learning Aksel Kretsinger-Walters, adk2164
Actor-Critic (Lec5) Summary adk2164@columbia.edu
ORCS 4529: Fall 2025 **Due:** — 11:59 PM ET

Lecture 4 Summary: Exploration–Exploitation Theory

Overview

The exploration–exploitation dilemma is central to reinforcement learning: how to balance gaining new information (exploration) versus maximizing known rewards (exploitation). This lecture formalizes that tradeoff through the *multi-armed bandit (MAB)* framework and explores provably optimal algorithms for regret minimization.

1. The Multi-Armed Bandit (MAB) Problem

A K-armed bandit has K independent actions ("arms"). At each time $t = 1, 2, \ldots, T$:

- The learner selects an arm $a_t \in \{1, \ldots, K\}$.
- Receives a random reward $r_t \sim \nu_{a_t}$ with mean μ_{a_t}.

Goal: maximize cumulative reward or equivalently minimize *regret*:

$$R_T = T\mu^* - \sum_{t=1}^{T} \mu_{a_t}, \qquad \mu^* = \max_i \mu_i.$$

Regret quantifies the cost of exploration—how much reward is lost compared to always pulling the best arm.

2. Naïve Approaches

Greedy Algorithm. Always choose the arm with highest sample mean $\hat{\mu}_i$. Fails because it may prematurely exploit a suboptimal arm due to early noise.

ϵ-Greedy. With probability ϵ_t, explore randomly; otherwise exploit:

$$a_t = \begin{cases} \text{uniform random arm}, & \text{with prob. } \epsilon_t, \\ \arg\max_i \hat{\mu}_i, & \text{with prob. } 1 - \epsilon_t. \end{cases}$$

Typically ϵ_t decays over time (e.g., $\epsilon_t = 1/t$). Simple, but not optimal.

3. Optimism in the Face of Uncertainty

A more principled strategy: select arms using **Upper Confidence Bounds (UCB)**.

UCB1 Algorithm (Auer et al., 2002). At time t, choose

$$a_t = \arg\max_i \left(\hat{\mu}_i + \sqrt{\frac{2 \log t}{n_i}} \right),$$

where n_i is the number of times arm i has been pulled.

This embodies the idea of *optimism under uncertainty*: each arm's estimate is inflated by a confidence bonus that decreases as n_i grows.

Regret Bound.

$$\mathbb{E}[R_T] = O\left(\sum_{i:\Delta_i > 0} \frac{\log T}{\Delta_i} \right), \qquad \Delta_i = \mu^* - \mu_i.$$

This is provably near-optimal (Lai & Robbins, 1985).

4. Thompson Sampling (Bayesian Bandit)

A Bayesian alternative maintains a posterior $p(\mu_i|\text{data})$ for each arm:

(a) Sample $\tilde{\mu}_i \sim p(\mu_i|\text{data})$ for all i.

(b) Choose $a_t = \arg\max_i \tilde{\mu}_i$.

(c) Update posterior with observed reward.

This balances exploration and exploitation automatically, achieving the same order of regret as UCB.

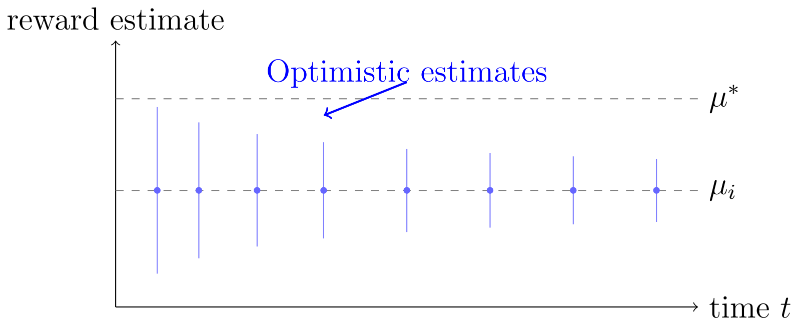

Figure 1: Illustration of UCB exploration: each arm's reward estimate is augmented by a shrinking confidence interval.

5. Lower Bounds on Regret

Lai & Robbins Bound (1985). Any consistent algorithm satisfies:

$$\liminf_{T\to\infty} \frac{\mathbb{E}[R_T]}{\log T} \geq \sum_{i:\mu_i<\mu^*} \frac{\Delta_i}{D_{\mathrm{KL}}(\nu_i \,\|\, \nu^*)}.$$

No algorithm can achieve asymptotically smaller regret.

6. Beyond the Classical Bandit

Contextual Bandits. At each step t, an observable context x_t is given, and rewards depend on both x_t and chosen action a_t. Policies $\pi(a|x)$ map contexts to actions.

Linear Bandits. Rewards follow $r_t = x_t^\top \theta^* + \eta_t$ with $\|\theta^*\|_2 \leq 1$. The OFUL (Optimism in the Face of Uncertainty for Linear models) algorithm chooses

$$a_t = \arg\max_a \left(x_{a,t}^\top \hat{\theta}_t + \alpha_t \|x_{a,t}\|_{V_t^{-1}} \right),$$

where $V_t = \sum_{\tau<t} x_{a_\tau,\tau} x_{a_\tau,\tau}^\top + \lambda I$. Regret scales as $O(d\sqrt{T \log T})$ for d-dimensional features.

7. Key Takeaways

- Exploration is essential to learn accurate estimates; too little causes bias, too much causes inefficiency.
- UCB provides a frequentist confidence-based approach with logarithmic regret.
- Thompson sampling provides a Bayesian counterpart with similar guarantees.
- Contextual and linear bandits generalize exploration–exploitation to structured, high-dimensional settings.